

VPA A-Level CS — AS Level Semester 1 Assessment Pack

AS Level Semester 1 — Assessment Pack

Complete Assessment Materials for Weeks 1 - 18 第1-18周完整评估材料

Assessment Philosophy: Regular, low-stakes formative assessment builds confidence; summative assessments mirror CIE conditions. All assessments include bilingual instructions and differentiated expectations.

评估理念：定期的低风险形成性评估建立信心；总结性评估模拟CIE考试条件。所有评估包含双语说明和分层期望。

1. Assessment Calendar

Week	Assessment	Type	Duration	Marks	Weighting
1	Baseline Diagnostic	Diagnostic	30 min	30	Formative
3	Information Representation Quiz	Quiz	20 min	35	Formative
6	Mid-Unit Test (Communication)	Summative	45 min	40	10%
9	Half-Term Examination	Summative	60 min	35	15%
12	System Software Essay	Summative	20 min	9	5%
15	Security Case Study Test	Summative	30 min	25	10%
18	End-of-Semester Mock	Summative	90 min	75	25%
Ongoing	Lesson Exit Tickets	Formative	5 min	n/a	Formative
Ongoing	Homework Tasks	Formative	30 min	n/a	5%

2. Baseline Diagnostic Test (Week 1)

AS Level Computer Science — Baseline Diagnostic

Time: 30 minutes | Marks: 30 | Instructions: Answer ALL questions. Show all working. Calculators permitted.

说明：回答所有问题。展示所有计算过程。允许使用计算器。

Section A: Multiple Choice (10 marks)

Q1. What is the denary equivalent of binary 10110110?

A) 182 B) 180 C) 178 D) 176

[1 mark]

Q2. Which of the following is a valid IPv4 address?

A) 256.100.50.25 B) 192.168.1.1 C) 192.168.1 D) 300.200.100.50

[1 mark]

Q3. What is the primary function of the ALU?

A) Store data B) Perform calculations and logical operations C) Control data flow D) Manage memory

[1 mark]

Q4. Which storage type has no moving parts?

A) HDD B) DVD C) SSD D) Magnetic tape

[1 mark]

Q5. What does DNS stand for and what does it do?

A) Digital Network System — connects computers B) Domain Name System — translates domain names to IP addresses C) Data Name Service — stores website data D) Dynamic Network Service — allocates IP addresses

[1 mark]

Q6. Which of the following is lossy compression?

A) PNG B) FLAC C) ZIP D) JPEG

[1 mark]

Q7. In the FDE cycle, which register holds the address of the next instruction?

A) CIR B) MAR C) PC D) MDR

[1 mark]

Q8. Which protocol is used to send email?

A) POP3 B) IMAP C) SMTP D) FTP

[1 mark]

Q9. What is the maximum positive number in 8-bit two's complement?

A) 255 B) 128 C) 127 D) 256

[1 mark]

Q10. Which of the following is an example of social engineering?

A) Brute force attack B) Phishing email C) DDoS attack D) SQL injection

[1 mark]

Section B: Short Answer (20 marks)

Q11. Convert the following:

- (a) Denary 217 to binary [2 marks]
- (b) Binary 11001101 to hexadecimal [2 marks]
- (c) Hexadecimal 3F to denary [2 marks]

Q12. A bitmap image has dimensions 800×600 pixels with 24-bit colour depth.

- (a) Calculate the file size in kilobytes. Show your working. [3 marks]
- (b) Explain why the actual file size might be smaller than your calculation. [2 marks]

Q13. Describe two differences between RAM and ROM. [4 marks]

Q14. Explain why packet switching is used on the Internet rather than circuit switching. [3 marks]

Q15. A student writes: "The control unit performs all calculations in the CPU." Is this correct? Explain your answer. [2 marks]

Mark Scheme

Q	Answer	Marks
1	A (182)	1
2	B (192.168.1.1)	1
3	B (Perform calculations and logical operations)	1
4	C (SSD)	1
5	B (Domain Name System — translates domain names to IP addresses)	1
6	D (JPEG)	1
7	C (PC)	1
8	C (SMTP)	1
9	C (127)	1
10	B (Phishing email)	1
11a	11011001 ($128+64+16+8+1 = 217$)	2
11b	CD (1100=C, 1101=D)	2
11c	63 ($3 \times 16 + 15 = 63$)	2
12a	$800 \times 600 \times 24 = 11,520,000$ bits = 1,440,000 bytes = 1,406.25 KB (accept 1,440 KB if shown in bytes)	3

Q	Answer	Marks
12b	Compression applied; metadata smaller than calculated; colour palette optimisation	2
13	RAM: volatile, read/write, stores running programs; ROM: non-volatile, read-only (mostly), stores firmware. Any 2 valid differences with explanation.	4
14	Efficient bandwidth use; resilience (packets can reroute); multiple users share lines; no dedicated path needed. Any 2 explained.	3
15	No; ALU performs calculations; CU coordinates and decodes instructions.	2

Diagnostic Analysis Guide

Score	Interpretation	Action
25 – 30	Strong IGCSE foundation	Gold tier; extension work from Week 1
18 – 24	Secure foundation, some gaps	Silver tier; targeted review of weak areas
Below 18	Significant gaps	Bronze tier; intensive IGCSE review; consider intervention

3. Information Representation Quiz (Week 3)

Information Representation Quiz

Time: 20 minutes | Marks: 35

- Q1. Convert denary 189 to 8-bit binary. [2 marks]
- Q2. Convert binary 11101011 to hexadecimal. [2 marks]
- Q3. Represent -45 in 8-bit two's complement. Show your working. [3 marks]
- Q4. Explain the difference between ASCII and Unicode. Why is Unicode necessary? [4 marks]
- Q5. A sound file is recorded at 44.1 kHz, 16-bit, stereo, for 4 minutes.
 - (a) Calculate the file size in megabytes. Show your working. [4 marks]
 - (b) Explain how the file size would change if recorded at 22.05 kHz instead. [2 marks]
- Q6. A company stores product images on its website. Explain why lossy compression might be appropriate. [3 marks]

Q7. Run-length encode the following data: AAAAABBBBBBCCCCDDDD [2 marks]

Q8. Explain one advantage and one disadvantage of using BCD instead of pure binary for financial calculations. [4 marks]

Q9. A bitmap image is 1024×768 with 32-bit colour depth.

(a) Calculate uncompressed file size in MB. [3 marks]

(b) The compressed file is 0.5 MB. Calculate the compression ratio. [2 marks]

Q10. Explain why the sample rate must be at least twice the highest frequency in the original sound (Nyquist theorem). [4 marks]

Mark Scheme (Summary)

Q1: 10111101 (2) | Q2: EB (2) | Q3: 11010011 (3) | Q4: ASCII=7-bit/128 chars, Unicode=universal/more bits, needed for world languages (4) | Q5a: $44100 \times 16 \times 2 \times 240 = 338,688,000$ bits = 42.336 MB (4) | Q5b: Halved (2) | Q6: Smaller files, faster loading, quality loss acceptable for web (3) | Q7: 5A5B5C4D (2) | Q8: Adv: exact decimal representation; Disadv: less efficient storage (4) | Q9a: 3 MB (3) | Q9b: 6:1 (2) | Q10: Prevents aliasing, ensures accurate reconstruction (4)

4. Mid-Unit Test — Communication (Week 6)

Communication & Networks Mid-Unit Test

Time: 45 minutes | Marks: 40

Q1. Distinguish between a LAN and a WAN. Give one example of each. [4 marks]

Q2. Compare client-server and peer-to-peer network models. Include two advantages of each. [6 marks]

Q3. Describe the packet switching process. Explain why it is suitable for Internet communication. [6 marks]

Q4. (a) State the function of a router. [2 marks]

(b) Explain why a switch is preferred over a hub in a modern LAN. [3 marks]

Q5. Draw and label the four layers of the TCP/IP protocol suite. Give one example protocol for each layer. [8 marks]

Q6. (a) Explain why IPv6 was developed. [3 marks]

(b) Describe two differences between IPv4 and IPv6 addresses. [4 marks]

Q7. Explain the role of DNS in Internet communication. [4 marks]

Mark Scheme Guidance

Q1: LAN=local/high speed; WAN=wide/span cities (2+example 1 each) | Q2: Client-server=central/server provides; P2P=no central/all equal; 2 adv each (6) | Q3: Split into packets/header+payload/reassemble at destination/rerouting possible; suitable=efficient/resilient/shared (6) | Q4a: Forwards packets between networks using IP addresses (2) | Q4b: Switch sends to correct port only; hub broadcasts; more efficient/secure (3) | Q5: Application(HTTP/FTP/SMTP), Transport(TCP/UDP), Internet(IP), Link(Ethernet/WiFi); diagram + examples (8) | Q6a: IPv4 exhaustion/4.3 billion insufficient/more devices (3) | Q6b: IPv4=32-bit/dotted decimal; IPv6=128-bit/hexadecimal; any 2 differences (4) | Q7: Translates domain names to IP addresses/hierarchical/root-TLD-domain/eliminates need to remember numbers (4)

5. Half-Term Examination (Week 9)

AS Level Computer Science — Half-Term Examination

Time: 60 minutes | Marks: 35 | Paper Style: CIE 9618 Paper 1 Section A

期中考试: 模拟CIE 9618试卷1 A部分风格

Section A (20 marks)

- Q1. (a) Convert denary 237 to 8-bit binary. [2 marks]
- (b) Convert the binary answer to hexadecimal. [1 mark]
- (c) Represent -100 in 8-bit two's complement. [2 marks]
- Q2. A digital photograph has dimensions 3000×2000 pixels with 24-bit colour depth.
- (a) Calculate the uncompressed file size in megabytes. [3 marks]
- (b) The image is saved as JPEG at 2 MB. Explain why this is possible and state one disadvantage. [3 marks]
- Q3. A 3-minute stereo audio file is recorded at 48 kHz with 24-bit sampling.
- (a) Calculate the file size in megabytes. [3 marks]
- (b) Explain how lossy compression could reduce this file size. [2 marks]
- Q4. Identify the network device that:
- (a) Connects different networks using IP addresses [1 mark]
- (b) Learns MAC addresses to send data to specific ports [1 mark]
- (c) Translates domain names to IP addresses [1 mark]
- Q5. Explain the difference between lossy and lossless compression. Give one example of each. [4 marks]

Section B (15 marks)

- Q6. Describe the packet switching process. Include the structure of a packet in your answer. [5 marks]
- Q7. (a) Draw a diagram showing the Fetch-Decode-Execute cycle. Label the registers involved. [5 marks]
- (b) Explain the role of the Memory Address Register (MAR) during the fetch stage. [2 marks]
- (c) State what happens to the Program Counter during the fetch stage. [1 mark]
- (d) Explain the purpose of the accumulator. [2 marks]

Grade Boundaries (Indicative)

Grade	Mark Range	Descriptor
A	28 – 35	Excellent understanding; accurate calculations; clear explanations
B	23 – 27	Good understanding; mostly accurate; some detail missing
C	18 – 22	Satisfactory; basic understanding; some errors

Grade	Mark Range	Descriptor
D	13 – 17	Limited understanding; significant gaps
E	8 – 12	Basic recall only; major misconceptions
U	0 – 7	Insufficient evidence of understanding

6. System Software Essay (Week 12)

System Software Essay Assessment

Time: 20 minutes | Marks: 9 | CIE-style 9-mark essay

Question:

"Compare paging and segmentation as memory management techniques. In your answer, include:

how each technique works

advantages of each technique

disadvantages of each technique

a conclusion stating which is more suitable for a general-purpose operating system."

[9 marks]

Mark Scheme (Levels of Response)

Level	Marks	Descriptor
4	8 – 9	Detailed comparison; clear explanation of both; valid advantages/disadvantages; reasoned conclusion; well-structured
3	6 – 7	Good explanation of both; some comparison; most advantages/disadvantages included; conclusion present
2	4 – 5	Basic explanation of both; limited comparison; some advantages/disadvantages; weak conclusion
1	1 – 3	Limited understanding; one technique described poorly; minimal comparison
0	0	No relevant content

Indicative Content

Paging: Fixed-size pages; OS maintains page table; no external fragmentation; internal fragmentation possible; simple to implement; fast allocation.

Segmentation: Variable-size logical segments; segment table; no internal fragmentation; external fragmentation; matches program structure; protection per segment.

Conclusion: Paging generally preferred for general-purpose OS due to simplicity, no external fragmentation, easier management. Segmentation better for protection/sharing.

7. Security Case Study Test (Week 15)

Security Case Study Assessment

Time: 30 minutes | Marks: 25

Scenario:

Victoria Park Academy stores student records on a central file server. The server is accessible to all staff via the school network. Students use the network for coursework. Last week, the IT department discovered that several files had been encrypted and a ransom note was displayed on the server.

场景：维多利亚公园学院将学生记录存储在中央文件服务器上。所有教职员工可通过学校网络访问该服务器。学生使用网络完成课程作业。上周，IT部门发现多个文件被加密，服务器上显示勒索信息。

Q1. Identify three different types of malware that could have caused this incident. For each, briefly explain how it operates. [6 marks]

Q2. Explain how the malware might have entered the school's network. Include two different methods in your answer. [4 marks]

Q3. The school is considering its response. Evaluate the following options:

(a) Pay the ransom [3 marks]

(b) Restore from backups [3 marks]

(c) Rebuild the server from scratch [3 marks]

Q4. Recommend technical, physical, and procedural security measures the school should implement to prevent future incidents. [6 marks]

Mark Scheme (Summary)

Q1: Ransomware (encrypts files/demands payment), Worm (self-replicates/spreads via network), Trojan (disguised as legitimate/creates backdoor). 2 marks each (type + explanation). | Q2: Phishing email (staff clicked link), infected USB (student plugged in), unpatched vulnerability, social engineering. Any 2 with explanation (4). | Q3a: May not get key/funds criminals/encourages attacks/no guarantee (3). | Q3b: Best option if backups exist/need recent backups/tested recovery/time to restore (3). | Q3c: Most secure/ensures clean system/loss of data if no backup/time consuming (3). | Q4: Technical (firewall, anti-malware, encryption, IDS), Physical (biometric access, CCTV, locked server room), Procedural (staff training, backup policy, incident response, password policy). Two from each category (6).

8. End-of-Semester Mock Examination (Week 18)

Time: 1 hour 30 minutes | Marks: 75 | Syllabus: 9618 Paper 1

AS水平计算机科学——试卷1模拟考试

INSTRUCTIONS TO CANDIDATES

Write your name, class, and candidate number in the spaces provided.

Answer ALL questions.

Write your answers in the spaces provided on the question paper.

Marks for each question are shown in brackets [].

Show all your working.

Calculators are permitted.

考生须知：回答所有问题。在试卷提供的空间内作答。展示所有计算过程。允许使用计算器。

Section A: Short Answer (25 marks)

Answer ALL questions.

1 (a) Convert the denary number 219 to 8-bit binary.

[2 marks]

(b) Convert your answer to part (a) into hexadecimal.

[1 mark]

(c) Represent the denary number -73 in 8-bit two's complement.

[2 marks]

2 A digital audio file is recorded with the following parameters:

Sample rate: 44.1 kHz

Bit depth: 16 bits

Channels: 2 (stereo)

Duration: 5 minutes

(a) Calculate the file size in megabytes. Show your working.

[4 marks]

(b) Explain how lossy compression could reduce this file size while maintaining acceptable quality.

[3 marks]

3 (a) Describe the difference between a bitmap image and a vector graphic.

[2 marks]

(b) A company logo needs to be displayed at different sizes on various media. Explain why a vector graphic would be suitable.

[2 marks]

4 Explain why Unicode is necessary instead of ASCII for worldwide communication.

[3 marks]

5 (a) State two advantages of using packet switching rather than circuit switching for Internet communication.

[2 marks]

(b) Describe the structure of a packet.

[3 marks]

6 Identify the network device that:

(a) forwards packets between networks using IP addresses [1 mark]

(b) learns MAC addresses to send data only to the intended recipient in a LAN [1 mark]

(c) translates domain names into IP addresses [1 mark]

7 Draw and label the four layers of the TCP/IP protocol suite. Give one example of a protocol found at each layer.

[8 marks]

Section B: Structured Questions (50 marks)

Answer ALL questions.

8 (a) Describe the purpose of each of the following registers in the CPU:

(i) Program Counter (PC) [1 mark]

(ii) Memory Address Register (MAR) [1 mark]

(iii) Current Instruction Register (CIR) [1 mark]

(iv) Accumulator (Acc) [1 mark]

(b) Describe the fetch stage of the fetch-decode-execute cycle. Include the role of registers in your answer.

[4 marks]

(c) Explain the purpose of the clock in the CPU.

[2 marks]

9 The following assembly language program uses a single accumulator architecture:

START: IN

STO 100

IN

STO 101

LDD 100

CMP 101

JPG GREATER

LDD 101

JMP OUTPUT

GREATER: LDD 100

OUTPUT: OUT

END

(a) State the purpose of this program. [1 mark]

(b) Complete the trace table for inputs 7 and 3:

Instruction	PC	Acc	Mem 100	Mem 101
START	1			

[5 marks]

(c) Explain the function of the **CMP** instruction in this program. [2 marks]

10 (a) Describe the difference between a compiler and an interpreter. [4 marks]

(b) Explain the purpose of lexical analysis during compilation. [3 marks]

(c) State one type of error detected during syntax analysis. Give an example. [2 marks]

(d) Explain why a two-pass assembler is needed rather than a one-pass assembler. [3 marks]

11 A hospital stores patient records on a computer network. The records contain sensitive personal and medical information.

(a) Explain two different types of malware that could threaten the hospital's data. [4 marks]

(b) Describe how encryption could protect the patient records. [3 marks]

(c) The hospital is considering using biometric authentication for staff accessing patient records. Discuss one advantage and one disadvantage of this approach. [4 marks]

(d) Explain the purpose of a digital certificate for the hospital's website. [3 marks]

12 "The development of artificial intelligence will cause more harm than good to society."

Discuss this statement. In your answer, you should consider:

benefits of AI development

risks and ethical concerns

your reasoned conclusion

[9 marks]

Mark Scheme (Indicative — Full scheme available in teacher pack)

Q	Max	Key Points
1	5	11011011, DB, 10110111
2	7	$44100 \times 16 \times 2 \times 300 = 423,360,000$ bits=52.92 MB; psychoacoustic models remove inaudible sounds
3	4	Bitmap=pixels/lossy scale; Vector=math/scalable
4	3	ASCII=128 chars/English only; Unicode=universal/CJK/supports all languages
5	5	Efficient use/resilience/shared lines; Header+Payload+Trailer with source/destination/sequence

Q	Max	Key Points
6	3	Router, Switch, DNS server
7	8	4 layers named + 4 protocols correctly matched
8	10	Register functions; fetch stage with MAR/MDR/PC/CIR; clock synchronisation
9	8	Finds maximum of two numbers; trace table accurate; CMP sets flags for conditional jump
10	12	Compiler vs interpreter (speed/debugging/portability); lexical analysis tokens; syntax error example; forward references
11	14	Two malware types explained; encryption process (plaintext→ciphertext→key); biometric adv/disadv; certificate identity verification
12	9	Benefits (efficiency/medical/transport); risks (bias/jobs/privacy/weapons); reasoned conclusion

Grade Boundaries (Indicative)

Grade	Raw Marks	UMS
A	60 – 75	80 – 100%
B	50 – 59	70 – 79%
C	40 – 49	60 – 69%
D	30 – 39	50 – 59%
E	20 – 29	40 – 49%
U	0 – 19	Below 40%

9. Assessment Administration

Before Each Assessment:

- ☐ Seating plan prepared (spread students apart)
- ☐ Papers printed and checked for errors

- ☐ Extra paper and graph paper available
- ☐ Clock visible to all students
- ☐ Mobile phones collected
- ☐ Bilingual vocabulary list available on request (for EAL students, not in exam conditions)
- ☐ Access arrangements confirmed (extra time, reader, scribe as per JCQ regulations)

During Assessment:

- ☐ Start and end times recorded
- ☐ Students reminded of time remaining at 15 min and 5 min
- ☐ No talking permitted
- ☐ Toilet breaks supervised

After Assessment:

- ☐ Papers collected in order
- ☐ Marked within 1 week
- ☐ Feedback given within 2 weeks
- ☐ Marks recorded in assessment tracker
- ☐ Intervention identified for students below target

10. EAL Support in Assessments

Permitted Support (Formative only):

- ☐ Bilingual glossary sheet (provided by teacher, no definitions)
- ☐ Extra reading time (10% additional for timed assessments)
- ☐ Clarification of question wording (not content)

Not Permitted (CIE Exam Conditions):

- ☐ Dictionaries (unless access arrangement approved)
- ☐ Bilingual glossaries in formal mocks (to simulate real conditions, but provided for Weeks 1–15 assessments)

Preparation Strategy:

- ☐ Ensure all key terms taught bilingually before assessment
- ☐ Practise with CIE-style English questions throughout semester
- ☐ Build academic vocabulary progressively

11. Intervention Triggers

Trigger	Response	Responsible	Timescale
Below 50% in any summative assessment	1:1 diagnostic interview; targeted revision plan	Class teacher	Within 1 week
Two consecutive formative scores below target	Parent contact; after-school support session	Class teacher + HoD	Within 1 week
Homework not submitted 2+ times	Detention + catch-up task; parent contact	Class teacher	Within 3 days
End-of-semester mock below grade C	Intensive revision programme; consider exam entry	HoD + SLT	Before Semester 2
EAL student struggling with question comprehension	Extra vocabulary support; simplified question practice	EAL coordinator + Class teacher	Ongoing